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Author reply

Dear Editor:
We read Dr. Eisenberg's letter with the utmost care. As far as the items needing clarifications, we can offer the following considerations:

1. The aim of our study was to verify if patients who did not respond to latanoprost were responders to bimatoprost. The observed results offer a clear answer to this question. Discussing the biochemistry of bimatoprost is far beyond the aims of our study. However, while discussing our results, we carefully pointed out that both (1) poor esterification of the prodrug latanoprost to the active acid form and (2) differences in the involved receptors might offer an explanation for the observed phenomenon. Moreover, the characterization of bimatoprost as a prostamide was suggested by Woodward et al in 2001.[1] Another article produced by the same authors and offering more details has recently been published.[2]
2. We were quoting the results of a multicenter, prospective, randomized clinical trial that appeared in a peer-reviewed journal.[3] This procedure is, in our opinion, appropriate and correct and follows the rules of evidence-based medicine. We are well aware of the post hoc criticisms raised by Drs. Eisenberg, Toris, and Camras in a review published, under the sponsorship of Pharmacia, in a supplement of *Survey of Ophthalmology*.[4] The issue of the face-to-face comparison between latanoprost and bimatoprost is still a matter of debate[5,6] and is not pertinent to our nonresponders study.
3. Dr. Eisenberg states that "it is difficult to generalize the findings of this study to clinical practice" and "clinically there are patients who respond to latanoprost but do not respond to bimatoprost," and asks "What would a study of this group prove?" To the best of our knowledge, our study is the first one to give proper scientific support to the general anecdotal experience quoted by Dr. Eisenberg (i.e., "everyone who practices medicine encounters patients who respond in unusual ways"). In our opinion, this is the most correct way to move from evidence-b(i)ased medicine to evidence-based medicine. Finally, the big family of the so-called hypotensive lipids (i.e., travoprost, latanoprost, bimatoprost, unoprostone) may be rather unique in the heterogeneity of the response given by each individual patient to each individual drug: does it make sense, whenever a patient gives a suboptimal response to a member of this family, to shift to another member of the same family instead of shifting to another family of drugs? Our controlled clinical trial gives some scientific support to this hypothesis.

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MRI to Differentiate Causes of Optic Nerve Disease

Dear Editor:
Rizzo et al[1] reported a retrospective evaluation of the use of magnetic resonance imaging (MRI) to differentiate optic neuritis from nonarteritic anterior ischemic optic neuropathy (NAION). They found that 5 of 32 patients with NAION had abnormally short T_1 inversion recovery signals on MRI in the clinically affected eye and that 2 patients had enhancement of the optic nerve. We have previously reported optic nerve enhancement on MRI in arteritic ischemic optic neuropathy.[2] Although it is not customary to perform an MRI for NAION, many patients arrive in the neuroophthalmology clinic with the outside MRI in hand. It has been my clinical impression that NAION does not produce enhancement of the optic nerve. When I have seen enhancement of the optic nerve in patients with a clinical picture suggestive of NAION I have pursued a more aggressive evaluation for inflammatory optic neuropathy (e.g., sarcoid) and giant cell arteritis (i.e., temporal artery biopsy).

It has been my assumption that small-vessel central nervous system ischemia (e.g., the presumed ischemic subcortical and periventricular white matter lesions) does not produce enhancement but that demyelination (e.g., acute periventricular white matter lesions in multiple sclerosis) and vasculitis (e.g., giant cell arteritis) do produce enhancement. I wonder if the authors could comment on (1) the possibility of giant cell arteritis in the two cases they reported of presumed NAION and optic nerve enhancement and (2) the mechanism for the presence (or absence) of optic nerve enhancement in NAION.

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Author reply

Dear Editor:
With regard to Dr. Lee's question of a possible alternative diagnosis of arteritis in patients with enhancing magnetic resonance imaging lesions whom we presumed to have nonarteritic anterior ischemic optic neuropathy (NAION), we were mindful to exclude cases that were even suggestive of arteritis. Our description of methods was intended to make clear our attempt to identify and eliminate such cases.

We also were curious about the two cases of "NAION" that had enhancement of the optic nerve, which prompted us to review these cases again. Of the two patients with enhancement, neither reported symptoms consistent with giant cell arteritis or other causes of inflammatory or infectious optic neuropathy. The erythrocyte sedimentation rates were 7 and 22, respectively, and complete blood counts were normal. Angiotensin-converting enzyme, antinuclear antibody, and fluorescent treponemal antibody blood test results were normal in one patient (none of these were obtained in the other). Both patients had small cup/disc ratios. Follow-up examination did not uncover evidence of diagnoses more likely than NAION.

Interestingly, the magnetic resonance imaging scan of patient 56 was interpreted as normal by both the senior author and a neuroradiologist (other than the study neuroradiologist). Our study neuroradiologist (JDR) interpreted the scan as showing a small area of "slight enhancement" of the clinically affected optic nerve, which according to the design of the study was recorded as a positive scan. By coincidence, this scan was randomly selected for a re-review (by the same neuroradiologist) to determine the degree of consistency of interpretations, and the second interpretation was the same (i.e., slight enhancement). The variance among reviewers indicates that this abnormality was subtle and perhaps of questionable clinical significance.

We do not have an explanation for the apparent enhancement of the optic nerve on magnetic resonance imaging scans of patients who are presumed to have NAION. We accept the possibility that the radiologic findings might indicate an alternative diagnosis, although none has become evident in these two patients.

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Retinal Hemorrhages in Children

Dear Editor:
In Schloff et al's article,[1] all cases of suspected nonaccidental trauma were excluded. Because two of the authors in a previous article stated the finding of retinal hemorrhages in a child with a head injury suggests a nonaccidental cause,[2] the conclusion that retinal hemorrhage is uncommon in children with intracranial hemorrhage not resulting from shaken baby syndrome is unwarranted. The only trauma cases left in the study should have been those without retinal hemorrhage. Indeed, the 7-year-old should probably have been excluded, although the earlier article only studied children younger than 3. The correct conclusion is that very few children with retinal hemorrhage cannot be suspected as having been abused.

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Author reply

Dear Editor:
Although we welcome Dr. Gardner's letter, we found it difficult to interpret his comments. We think he is concerned that the children in our study were preselected in such a way as to eliminate those children with retinal hemorrhages who might have been assumed to have been victims of the shaken baby syndrome.

We did exclude all cases of suspected nonaccidental trauma. However, this was done by the admitting hospital service *before* the child was examined by the ophthalmology team. Therefore, retinal hemorrhages were not used to predetermine the diagnosis. Interestingly, those children who were excluded from our study based on an initial suspicion of possible abuse (all children had their eye examinations within 72 hours of admission) but whose cases were later adjudicated by our child abuse team as accidental injuries had no retinal hemorrhages. Had they been included in our study, this would have further underscored the rarity of retinal hemorrhage associated with intracranial hemorrhage in children who are not victims of shaken baby syndrome.

Dr. Gardner refers to another publication from our center.[1] Although the intent of that study was not to specifically address the issue of retinal findings in children with intracranial hemorrhage, of the 79 children with head injury, 8 had intracranial hemorrhage. Three of those eight cases were nonaccidental trauma, and all had retinal hemorrhages. The remaining five cases of accidental trauma all showed no evidence of retinal hemorrhages, again supporting our conclusion that "retinal hemorrhage is uncommon in children with intracranial hemorrhage" not due to child abuse.

We agree that the diagnosis of shaken baby syndrome